Meeting on Intelligence - Bethesda - Nov 30, 1960

Rich Ferrill - chairman

Purposes:

1. Identify weak spots in Intel. which IBM can attack
2. Identify specific programs to develop equipment & techniques.
3. Procedures/structure for team work.

(2 years) - what equip is coming -- what should be coming -- Kingston Engr

Ferrill -- Overview

agencies: "Wash. Community"

CIA, NSA, Army, Navy, AF, State, FBI, ACC

D/I JCS, Sec Defense.

Prime users of intel - unified & specified commanders

SAC, NORAD, PAC, MED, EUR, ALASKA, CARIB,

ATL FLEET, ATlant Europe

other: NATO
British

Field collectors:

Human, document, Photo, Electronic U-2, RB-47

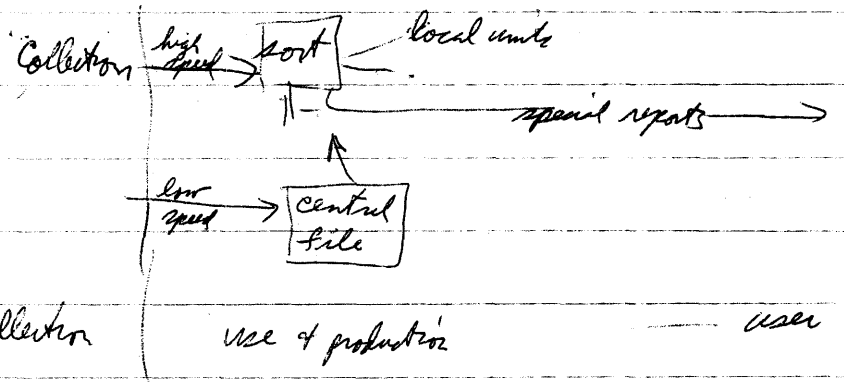
(4 M frames per year)

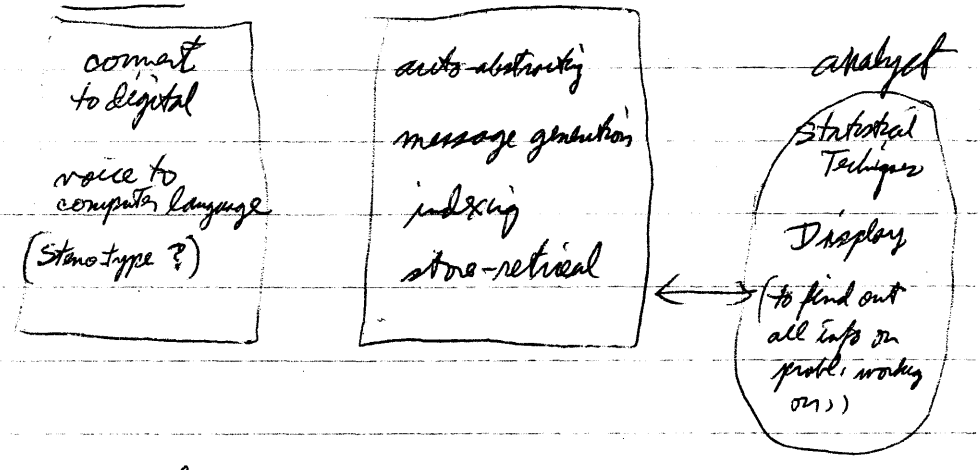
Volume, changing scenarios

Need to maximize -- far from it now,

avoid surprise

"a series of approximations in which the load is always increased" – Valley geo





Arlington Hall -- has 709 2 yrs ago.

now 7090

35 M being spent on new collection equip. alone

"sensible programs -- sensibly proposed"

long range potential -- with interim pay offs.

change of success -- must get politicians convinced

big gamble right around corner --

466L (Dunkin)

6 to 7 tons of paper per month (cards, tape, & paper)

Page reader
auto map compilation,

(Bob Cortney.) Reconnaissance

	aircraft	satellites	ground-based
ELINT	✓	✓ SAMOS	✓
COMINT	(—)	(—)	(—)
IR	✓	✓	?
RADINT	✓	✓	✓
Non-intentional Radiation	—	—	—
Photo,	✓	✓	

Subsystem I -- victim of over design & changing operational & functional - want redesign to meet recon SAMOS problem,

Subsystem F - ELINT.

Photo interpreter -- still must be done by

$200 M have gone into ELINT since 1949

$60 M to present program,

ADD-4 (1947) Poor system design - hard to put in aircraft
don't understand problem - too much black boxing
- no simulation was tried,

low volume procurement rates. (3 max of one machine) ($15.5M has gone into FINDER)

- very high data rates - low info = characteristics of RADINT, ELINT

ASD-1 eg, 1500 intercepts/sec for ASD-1
(3 36 bit wds ea)

(63 x 10^6 wds in 4 hours)

freq, pulse width, bearing, rep. rate

ALD-4 144,000 frames in 4 hour mission of CRT pictures,

---

we hope proposed 7074 plus 1401 to process such
- high speed simplex machine

a general purpose machine has trouble when 90% of data is bad, -- needs displays,

MELPAR: 2000 data in 5 colors displays for FINDER

Problem area → equipt. for Direction; finding one sensor with another
eg direct camera toward radar site from elint recon.

(Jack Stone) Intel Dept. - just starting Adv. techniques & sys. analysis
also look at new customers

"Division Coordinator - sort of" = Ferrill

exchange personal ≈ - watch ⟷ onsite

(C. Medow) Fleet Intel small systems -- as consultants to Navy can use 1401

- restrictions of money & manpower.
also could not stop present operations

Need: - a page reader badly needed to cut out key punching etc.
- a device to do indexing - a message composer? - a keyword selector?
- handling photo input (1M to 3M frames in library) - ~~cost~~
about ½M charge per year.
look for changes -- some analysis,
associate memory, -- (no details on size, etc.)

Task: - to keep up target folders - to get plans to target.
scheduled strikes -- pre planning subject to last min changes
called strikes ← done on carrier Rem Rand M460

HARPA
- some proj. analysis,
- systems & photo handling problems,
- air, analytic techniques eg. what would one want to use games?

---

another prob: Management science area -- getting info. in front of right guy
cut down duplication

HGK

intell – how much worth, can't taste, feel, it's hard to know how good. subs.

= users point of view

info retrieval problem
indexing,
displays

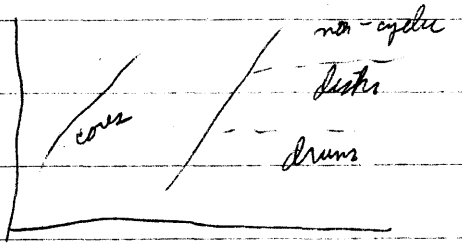
= evolutionary system – – have more power & flex. than
= decision maker not analyst.
= nothing succeeds like success

(Ehlers) goto source

- paper tape (5 chan)
- page printer – reader

approx at USIB level

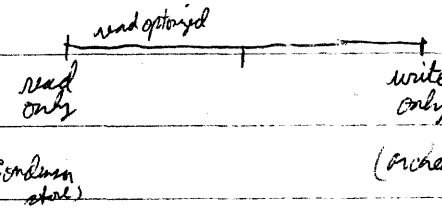
(Cypher) C.D. Jackson – mem capacity vs speed chart



core 10^{12} per bit

drums 1 ¢ per bit

disks 0.1 ¢ per bit



Ruggedized Low Cost File (58)

2×10^6 characters per disk pack (removable) can have four drives on one channel

200 ms max.
50 ms min.

- single disk file = more adv. state of art, + (same speeds)

(Behnke) Associative Memory

Random storage
Retrieval by content
High Storage utilization
Ease of File Maint
Operation Speed thru parallelism
Intermingled Logic & Memory

Range compare -

- very useful when compares are rare events

10,000 wd mem
42 bit/wd
1 cm ft wd.

high low equal
Range retrieval
sequential read out
selective write
Masking

load input regs	0.2 μs
compare	1.0 μs
read out	2.5 to 5.0 μs

Logic element -

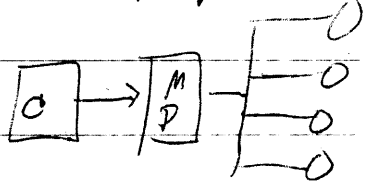
- comparison for like things in mem
- 3 D file display

(Ploughy)

nearest lower value search

- select L group
- interrogate with successive mask checking no. equals until only one matches.

(Bachelor) Displays (new & sys displ)



consoles

1800, Cometer alphanumeric flicker-free rate.

a complete display ~$100,000 cost
no regeneration needed

- has selection system.

about $2M has been put in
$100K to get going
150K to make modify to fit 7090 (orient to type channel)

10" screen 6" vectors.

geography -- presented at low flicker / per sec.

Message composer : ~$10,000 ea
Typewriter, (35,000 for 1 model)

Large Board graphic Display
2 - 3 sec update Time

Large Board alphanumeric Display, ~$10 000

multi terminal remote display -- low cost at terminal.

raster scan 2000 line resolution
½ sec. for each picture

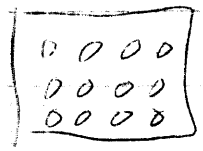
- high ambient light (Sam J.)

4 year - drawing reader & display (20×20' drawings or 35 mm film)
does tie up computer ~ reads line segments
4000, line resolution

- hard copy 8½×11 fractional sec. rates.

(The above will speed up only if specific customers pay to speed up.)

⇒ can draw picture of keyboard on screen



- can touch them & send signals back to computer

(Al Warhit) Information Retrieval

- Tendency to accept past methods & mechanics it,

methods: classification
avoid language problems
synonyms,
also can give hierarchies and

explicit & implicit info.

Thesaurus 6M organic compounds cataloged
"schlagwort" indexing
search vs look-up ← good for quick matching
File need not be organized

New subjects eg. { Psychology was originally Philosophy & Religion
was reclassified to Science 20 yrs later

in future - relativistic
statistical approaches. } easy for computers

Yale Univ. - research libraries would need 6000 catalogers.

"human interface" - OK as long as question is clear,
negotiate question,

2nd day: Intelligence Meeting Bethesda, Dec 1, 1960

(Geo Petrie)

story: San Antonio - Bill Brand - AF Future plans (7 years) -- most are still there haven't been done, diagram 3/4 of room

Walnut: 1st program in image technology
optical + photographic (Kalvar film)
ultra violet sensitive
heat fixed,

Resolution: 245 - 250 lines/mm

newer: 450 lines/mm have not been able to reproduce
(may be shelf life ?)

X-129 600 lines/mm 4 times slower

Planned Reduction: 40 to 50 (50 to 60) times in going to film,

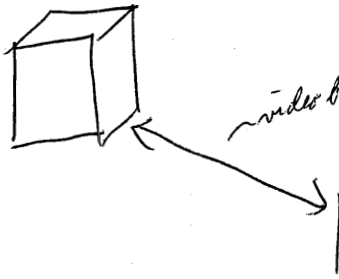
Problem: the copier very great deal in quality
carbons, offset typewriter, etc.
as

as result had to go to 35 diameters -- reduces capacity by factor of 2
990,000 images per module
instead of orig 2 million desired.

Minicard: has index on record. Walnut has separate index
CIA based on name index.

Teleprocessing went from ASD to DSD,
Example: American Airlines job

MIRAC: "Management Info. Retrieval and Communication"



Terminal ← being looked at SJ. Technical probs only. (feasibility hasn't been proved)

1. Video display
2. Hard copy input
3. Hard Copy output
4. Control -- to scan thru material (2 screens?)
5. Priority

Ref: Peter Luhn - routing of documents
Bob Tracedr (?) (SJ)

"Low Cost -- at any price",

- Security of some type of information.
- What kinds of questions would be asked of such a device?

Speech Recognition (SJ. device - recognizes spoken digits)

Examples: Weather prediction -- pictures from Satellites.

- When does one want to digitize analog document, & when does one want to keep analog record?
  example: engr. drawings -- can be digitized, 10-15 to 1 reduction
  (4000 characters/page) "standard page"

Subsystem I needed wider bandwidth -- too expensive & not secured -

compacting data before transmission.

ASD: Bob LaHaye (?) -- using known characteristics of printed pages -- also big drawings.

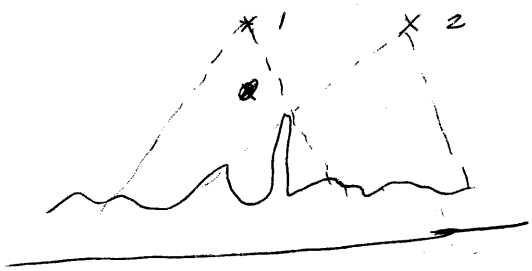
(Cliff Williams - Kingston) Image Proving -- Map Compilation,

Ft. Belvoir -- study to do map compilation digitally,

has contour

3D models

Tulane - Univ - tumor contours,



picture -- digitize 400 to 1000 spots per inch



can calculate altitude of each

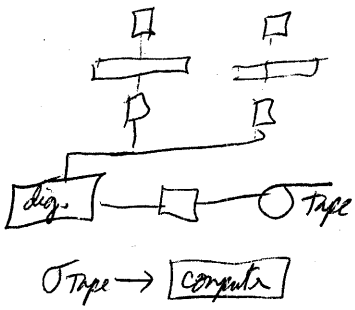
Shades of gray: black = 0

white = 7

can go to 170 shades of gray

(eye can do up to 100 by comparison
10 individually)

scan picture : 1 mil spot.



convert orthographically

Philosophy: get in digital form as soon as possible & keep it as long as possible,

pattern matching problem:

must take out tilt in picture

can't pattern match grays themselves -- must use differences,

compress places & minuses first — then value of nos — then distances

Example of automobiles at corner,

Don Sanctis -- has done computer

$4\frac{1}{2}$" × 9" 15M × 3bits = 45M bits

scan 15 min
compare 15
compute 1^h30^m 70¢ — all conditions (3½ months to do whole job by hand)

facsimile 400 spots/inch ← (4½ feet on ground at 20,000 feet.)
50 spots/inch = newspaper. (¼ mile square from satellite)

(one picture has ~ 500 reels of tape)

can compress by transmitting { shades & no. the same -- 1st difference
not same changes 2nd difference

CBS photo scan tube,

Can get oblique views - perspective,

(Joe —) Statistical Techniques

Discriminant Function

Populations: $\Pi_1, \cdots \Pi_p$

observations: $(X_1, X_2 \cdots X_n)_t$ known which Π they belong to.

a new observation: which class is it in?

~~alert status corresponding to Π's~~

Example: Word frequency class re warning

Problem: ie, propaganda broadcasts to raise public's desire for war.

keywords.

---

assume: attitude of Russian $A_1, A_2 \cdots A_a$

alert status $A_1^* \cdots A_a^*$

key words $W_1 \cdots W_a$

and assume rates of key words monotonic increasing or decreasing.

$$r_{1k} \geq r_{2k} \cdots \geq r_{ak}$$

or $\leq \quad \leq$

Data screening problem: - must estimate rates.

1. Sources $\{S_i\}$ must be periodic can be observed.
2. $C_t(S_i)$ condition of sources
3. Pick keywords $\{W_i\}$ -- take large set + reduce later, judgement (or phrases)
4. assign time interval δ_W
5. obtain freq. count $f_t(W_i, C_t(S_j))$ from sources as fn time

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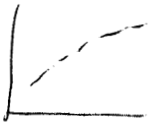
to estimate rates: (t, f_t)

$$f_t = \sum_{i=1}^{K} \gamma_i f_i(W_j, C(S_i))$$

weight functions γ_i

next order in time for all keywords. $(t_1 f_1), (t_2 f_{t_2}) \cdots$

now sum: to get rate estimate $\hat{r}_{t^*} = \sum_{t^* \delta_w} f_t$ for $t^* - \delta_w \leq t < t^*$

can plot rates vs time 

Now compare rates to Population classes,

For 2 populations -- make decisions -- pairwise

let program build up from historical data

	t_1^*	t_2^*	t_3^* ...	t_n^*
w_1	$\hat{r}_{11}$	$\hat{r}_{12}$	...	$\hat{r}_{1n}$
⋮	⋮	⋮		
w_w	r_{1w}	...		
alert status	A_{21}^*	A_{22}^*	...	A^*

now see which w's should be used to give relationship

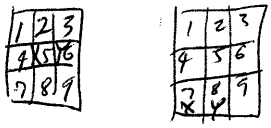
assign cost

- moment computations
- normal -- non normal --- skew
- categorical data

⇒ Such statistical techniques in many intel areas

(F. P. Brooks) Streaming Method,

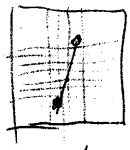
example: picture recognition



compare each 2 element sequences stored in row order. --

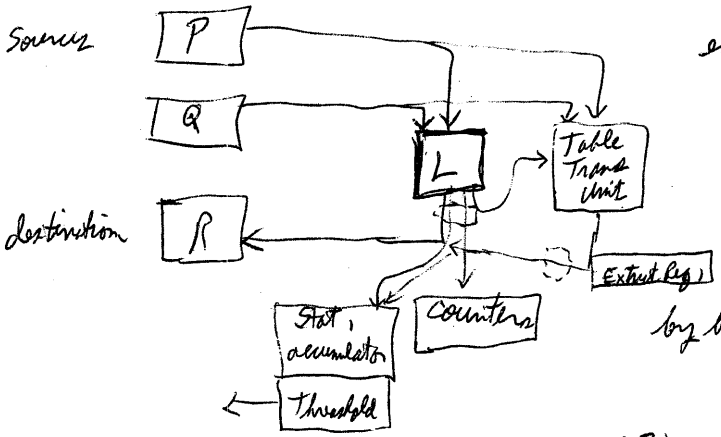
- can compare 2 by 2,
- use of shifting registers - goes faster -- form counts.

example: line of sight problem: given altitudes at given points,



- must skip certain values

imp → basic operation can be done on small amt of data at a time most of time, -- essentially a playboard, --- repeated many times,
- accumulate data - counts, - no fetching of instructions.



example --- --- ×
take 3 skip 4 take 3 | end rec.

built into hardware ¼ μs per byte

by byter or groups of bytes.

TRU can form address by P + Q + base addr. with offsets

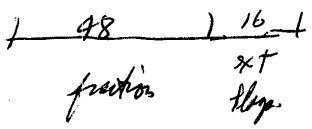
20 words of instructions - to set up stream

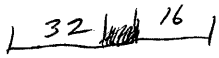
stopping conditions: can stop on output of L or E or on counts or accum., or indexing bounds

- Takes lot of hardware
- hard to program.

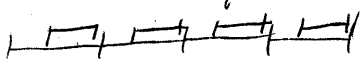
- smaller computer using streaming type (at 2 μs/byte)

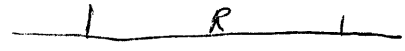
10AB: 64 bit wd, 16 4 bit byte, 8 bit bytes. { 1.8 × 709 scientific; 2.03 × 7070 commercial

- fl. pt: 

or 

- multiple fields: format relative to def. ie same . . do pairwise arithmetic on vector type operands.

eg 

- long record operations: 

(record may not be contin but will be treated as if continuous,

address given control word - giving starting address + length.

scatter - read write
- alphanumeric comparison,

long record opr { Transmit -- can specify transmit 'till match, or 'till dash; Compare; Connect; Translate (streaming table lookup)

3 kinds of table-look-up (like 709 convert)
1 - each byte gives result byte
2 - result byte + carry byte (left)
3 - result byte + carry right

afternoon

Tentative goals:

	Due Date
1. Test Model for Auto-Target Identification via photos (pattern matching)	1 year
2. Prototype Associative-Memory (~~for target selection~~ intel appls.)	2-3 yrs.
3. Prototype Auto-abstracting INDEX	6 mos.
4. Language Translation (IOC)	6 mos
5. Initial 438L Display	1 year
6. Page Reader	2 years.

Areas

1. Photo processing
2. Memories
3. Displays
4. Communications
5. DP Machine Compatibility
6. Language Translation
7. Analytic Techniques

±1. - programming support for Photo problem. FSD prog. - not great knowledge funding?

{ Cypert Kemill to spell out program not later than Jan 61

2. assoc. Mem,

how large ? not for targeting.

(2500.0 Target list – worth, vulnerability, location, etc.

→ Study to determine application – 3 months (country, ethnic)
will assemble ad hoc group

- size
- identity only ?
- logic on descriptor
- tag only ?

due date 2 or 3 year

3. Auto abstracting - Indexing of what ?

– a very important key for future application
--- Petrie has objections ---

→ System Center set up a seminar to learn State of art.

4. Language Translation.

Howerton's channel -- not now open ?
(40,000 pages a day in 6 mos ?)
need to look into Techniques

Rose disk - approach uses only one Theory of Translation.

→ C.R. Green will look into mkt problems.

J. Stone - requirements
(committee to be set up)

– can do on general purpose computer now within limited vocabulary –

photostore vs ADF ?

– need more info on requirement

5. Display Capability:

-- already too late for 438L,

-- product planing — marketing

→ Prod. Planing will evaluate, inputs within FSD

6. Page Reader;

Part FSD, part GP proposal ? — what is status

Baird Atomic has active program

7. Analytic Techniques

-- data compression,

→ another meeting to discuss longer range reqs.

Date: November 22, 1960
From (Dept/Loc): 500/Kingston Eng. Lab.

IBM

Subject and/or Reference: Intelligence Seminar, Nov. 30 & Dec. 1 (at Bethesda)

To: Mr. R. Farrell
IBM-FSD, Washington, D.C.

Presentations by your associates describing present and projected intelligence problems will be followed by presentations on relevant techniques and equipments. Where possible, projections of developments will be encouraged. Discussion will be encouraged in an attempt to highlight key problems, technological limitations and gaps in our programs. The following individuals plan to attend and present information and views on the subjects listed:

C. D. Jackson	FSD	Standard Memory Projections
A. Plonsky & F. Behnke	FSD	Associative Memories
W. L. Batchelor	FSD	Displays
G. Petrie	ASDD	Image Retrieval (Mirac)
C. F. Williams	FSD	Image Scanning & Processing
F. P. Brooks	DSD	Machine Organization
J. Warheit	ASDD	Information Retireval
C. Kossack	Research	Statistical Approaches
S. Campbell	DSD	Streaming Techniques

Several additional members of the High Speed Computational Techniques contract will also attend. All discussions will be classified SECRET.

R. J. Cypser

RJC:lms

cc: All Participants
Dr. E. Adams
Mr. H. E. Cooley
Mr. T. C. Cox
Mr. N. P. Edwards
Mr. A. W. Lampe
Mr. J. D. Lee
Mr. J. D. Newton
Mr. R. L. Rockefeller
Mr. L. R. Walters